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~~a) I~~ $F(c_1, c_2) = u(c_1) + \beta u(c_2)$ $u_c = \frac{c^{-\sigma}}{1-\sigma} \Rightarrow u'_c = c^{-\sigma}$

II) Consumption constraint, budget constraint

i)

$$c_1 + a = a_0 + y_1 - T_1$$

$$a = a_0 + y_1 - T_1 - c_1$$

ii) $c_2 = y_2 - T_2 + (1+n)a$

$$c_2 = y_2 - T_2 + (1+n)(a_0 + y_1 - T_1 - c_1)$$

$$\frac{\partial c_2}{\partial c_1} = -(1+n)$$

iii) $\frac{\partial F}{\partial c_1} = u'(c_1) + \beta u'(c_2) \cdot \frac{\partial c_2}{\partial c_1}$

$$c_1^{-\sigma} + \beta c_2^{-\sigma} [-(1+n)] = 0$$

$$c_1^{-\sigma} = \beta c_2^{-\sigma} (1+n)$$

$$(c_2/c_1)^{\frac{1}{\sigma}} = \beta(1+n)$$

iv)

$$c_2 = [\beta(1+n)]^{\frac{1}{\sigma}} \cdot c_1$$

$$y_2 - T_2 + (1+n)(a_0 + y_1 - T_1 - c_1) = [\beta(1+n)]^{\frac{1}{\sigma}} \cdot c_1$$

$$[\beta(1+n)]^{\frac{1}{\sigma}} + (1+n) \mid c_1 = y_2 - T_2 + (1+n)(a_0 + y_1 - T_1)$$

$$c_1 = \frac{y_2 - T_2 + (1+n)(a_0 + y_1 - T_1)}{[\beta(1+n)]^{\frac{1}{\sigma}} + (1+n)}$$

$$c_2 = [\beta(1+n)]^{\frac{1}{\sigma}} \cdot c_1$$

a?

$$I) C_1 = \frac{y_2 - T_2 + (1+n)(a_0 + y_1 - T_1)}{\left\{ [\beta(1+n)]^{1/\sigma} + (1+n) \right\} = \theta}$$

$$II) \frac{C_1}{y_1} = \frac{y_2/y_1}{\theta} - \frac{T_2/y_1 + (1+n)(a_0/y_1 - T_1/y_1 + 1)}{\theta}$$

III) C_1/y_1 AUMENTA COM A RAZÃO $\frac{1}{\theta}$ PARA AUMENTO DE y_2/y_1

IV) LOGO O AUMENTO DA EXPECTATIVA DE y_2/y_1 O CONSUMO AUMENTA PELA RENDA.

$$II) PARA $y_2 = T_1 = T_2 = 0$$$

$$C_1 = \frac{(1+n)(a_0 + y_1)}{\left\{ [\beta(1+n)]^{1/\sigma} + (1+n) \right\}}$$

$$II) \frac{\partial C_1}{\partial n} = \frac{(a_0 + y_1)}{\theta} \left[1 - \frac{1/\sigma}{\theta} \right] = \frac{(a_0 + y_1)}{\theta} \left[1 - \frac{[\beta(1+n)]^{1/\sigma}}{\theta} + (1+n) \right]$$

$$\frac{\partial C_1}{\partial n} = \frac{(a_0 + y_1)}{A} \frac{[\beta(1+n)]^{1/\sigma}}{\theta} [1 - 1/\sigma]$$

$$I) PARA $1 - 1/\sigma > 0$$$

$$(\sigma > 1)$$

O CONSUMO DIMINUI

$\sigma = 1$ É CONSTANTE

$\sigma < 1$ AUMENTA

$$1) \frac{\partial c_1}{\partial \tau_1} = -\frac{(1+n)}{[\beta(1+n)]^{1/6} + (1+n)} \quad ; \quad \frac{\partial c_2}{\partial \tau_1} = [\beta(1+n)]^{1/6} \cdot \frac{\partial c_1}{\partial \tau_1}$$

$$\frac{\partial c_1}{\partial \tau_2} = \frac{-1}{[\beta(1+n)]^{1/6} + (1+n)} \quad ; \quad \frac{\partial c_2}{\partial \tau_2} = [\beta(1+n)]^{1/6} \cdot \frac{\partial c_1}{\partial \tau_2}$$

ou?

$$2) c_1 = \frac{y_2 + (1+n)(a_0 + y_1) - (1+n)(\tau_1 + \tau_2/(1+n))}{[\beta(1+n)]^{1/6} + (1+n)}$$

TODA DEPENDÊNCIA COM TAXAÇÃO VEM:

$$\Delta = \tau_1 + \tau_2/(1+n)$$

$$\Delta' = \tau_1 - 1 + [\tau_2 + (1+n)]/(1+n) = \tau_1 + \tau_2/(1+n) = \Delta$$

LOGO ESSA MUDANÇA NÃO IMPACTA.

SIM, O TIMING IMPACTA NO CONSUMO COMO

VISTO NO ITEM D

2) a) ~~✓~~ SIGNIFICA O LIMITE DE ENDIVIDAMENTO

b) ~~✓~~ USANDO:

$$a = a_0 + y_1 - T_1 - c_1$$

$$a = \frac{-y_2 + T_2 + [B(1+n)]^{1/5} (a_0 + y_1 - T_1)}{[B(1+n)]^{1/5} + (1+n)} \geq -b$$

RENDA FUTURA MUITO ELEVADA, FARIA SE ENDIVIDAR PARA CONSUMIR MAIS NO PRESETE, PORÉM A CAPACIDADE DE ENDIVIDAMENTO É LIMITADA

c) ~~✓~~ QUANDO A LIMITAÇÃO $a \geq -b$ NÃO RESTRINGE, SERÁ O MESMO DA QUESTÃO 1. ~~✓~~

$$c_1 = \frac{y_2 - T_2 + (1+n) (a_0 + y_1 - T_1)}{[B(1+n)]^{1/5} + (1+n)}$$

$$c_2 = [B(1+n)]^{1/5} \cdot c_1$$

$$a = \frac{-y_2 + T_2 + [B(1+n)]^{1/5} (a_0 + y_1 - T_1)}{[B(1+n)]^{1/5} + (1+n)}$$

±) PARA A RESTRIÇÃO

$$a = -b$$
$$c_1 = y_1 + b - T_1$$

$$c_2 = y_2 - T_2 - (1+n)b$$

$$d) \cancel{a} \quad a = a_0 + y_1 - T_1 - c_1$$

$$c_2 = y_2 + (1+n-T_2)a$$

II) MAX:

$$F(c_1, c_2) = U(c_1) + \beta U(c_2)$$

$$\frac{\partial F}{\partial c_1} = \frac{\partial U}{\partial c_1} + \beta \underbrace{\frac{\partial U}{\partial c_2} \cdot \frac{\partial c_2}{\partial c_1}}_{-(1+n-T_2)}$$

III)

$$c_1^{-\sigma} + \beta c_2^{-\sigma} [-(1+n-T_2)] = 0$$

$$(c_2/c_1)^{\sigma} = \beta (1+n-T_2)$$

$$c_2/c_1 = [\beta (1+n-T_2)]^{1/\sigma}$$

IV)